

Somanshu Mohapatra

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SKILLS

Technical: Python, C++, Java, JavaScript, SQL, Embedded C, MATLAB, Git, Docker.

AI/Vision: PyTorch, TensorFlow, YOLO, OpenCV, Machine Learning, Sensor Fusion.

Robotics: ROS2, MoveIt, Gazebo, Kinematics (DH), PID Tuning, SLAM, State Machines, MQTT.

Electronics: Arduino, Raspberry Pi, Embedded Linux, UART, I2C, SPI, Proteus.

Design: SolidWorks, Fusion 360, AutoCAD, LightBurn, ANSYS HFSS, 3D Printing, Rapid Prototyping.

Professional: Leadership, Team Collaboration, Technical Communication, Problem Solving, Time Management.

EXPERIENCE

AutomateX: Industrial Automation, Lovely Professional University

March 2025

- Programmed PLC ladder logic in TwidoSuite, cutting execution time and improving simulated process speed by 14 percent.
- Wired and calibrated 8 sensor inputs and actuators, cutting setup testing downtime by 10 percent.
- Wrote and tested interlocks in ladder logic to prevent mechanical jams and emergency stops.

PROJECTS

2.4GHz Microstrip Patch Antenna, ANSYS HFSS

October 2025

- Modeled a 2.4 GHz microstrip patch antenna for Wi-Fi and Bluetooth bands, hitting a -28 dB return loss.
- Tuned substrate thickness and feed line geometry in ANSYS HFSS, boosting bandwidth by 15 percent and peak gain to 2.8 dBi.
- Simulated S-parameters and 3D radiation patterns to verify impedance matching and antenna efficiency.

Dynamic Audio Monitoring and Narration, Raspberry Pi

May 2025

- Built a portable speech-to-text unit on Raspberry Pi 4B, lowering text output latency by 350 milliseconds.
- Streamed AssemblyAI speech APIs via WebSockets to an I2C SSD1306 OLED screen using Python multi-threading.
- Tested continuously over 100 benchmark runs with zero dropped frames or WebSocket disconnects.

3-DOF Robotic Arm, MATLAB, Arduino

April 2024

- Designed and 3D-printed a 3-DOF robotic arm, holding a 1:4 payload-to-weight ratio over 50 test cycles.
- Linked MATLAB to Arduino over serial UART to drive motor joints in real time.
- Tuned PID controller gains in MATLAB to reduce motor jitter and improve positioning accuracy by 10 percent.

ACHIEVEMENTS

Patent: Enhancing Food Accessibility and Energy Efficiency in Refrigeration

September 2025

Application No.: 202511085342 A. Designed an internal tray mechanism and airflow layout for domestic refrigerators.

CERTIFICATES

- Database Management System, NPTEL.

August 2025

- FusionCraft Robotics Workshop, Resolute Lab.

March 2024

EDUCATION

Lovely Professional University

August 2023 – Present

BTech Robotics and Automation. CGPA: 8.38/10.0.

Shri S.N. Siddeshwar Public School

March 2021 – May 2023

Senior Secondary Education (PCMB). Percentage: 85.6%.

Ryan International School

March 2020 – May 2021

Secondary Education. Percentage: 81.6%.